SCOREMAP: Structure-Aware Grading of Handwritten CS Answer Scripts

Abstract

SCOREMAP grades handwritten CS answers by converting detected answer regions into a typed evidence graph and executing rubric items over that graph. Instead of predicting marks from a black-box page embedding, the system aligns definition spans, algorithm steps, complexity expressions, code-like lines, and diagram regions to rubric items, then returns grounded partial credit with evidence overlays.

Introduction

Handwritten CS scripts mix prose, pseudocode, complexity notation, and diagrams. This makes generic OCR-plus-classifier pipelines brittle and hard to audit. SCOREMAP reframes the problem as document understanding plus structured scoring, with the rubric treated as an executable program over localized evidence units.

Related Work

Donut motivates OCR-free document understanding, LayoutLMv3 motivates joint text-image-layout reasoning, and Pix2Struct motivates question-conditioned structured prediction from page images. We use those papers as prior-work anchors and baseline inspirations rather than claiming a stronger backbone.

Dataset and Annotation Protocol

This deliverable includes a pilot benchmark of 18 question-level samples across 6 CS question archetypes and 3 writer styles. Each sample has region boxes, transcripts, region type labels, rubric hits, and evidence-to-rubric links. The split is writer-wise: writer01 train, writer02 val, writer03 test.

Method

The pipeline has four main stages: (1) typed evidence extraction from answer regions, (2) typed evidence graph construction with reading-order and adjacency edges, (3) executable rubric matching with type-specific heads for definitions, algorithm steps, complexity items, code items, and diagram items, and (4) constrained score aggregation with review flags for low-confidence awards.

Experimental Setup

We evaluate three variants: a generic scorer without type routing, a no-graph ablation without order or prerequisite constraints, and the full SCOREMAP pipeline. Metrics include MAE on marks, exact-score match, weighted kappa, rubric-item F1, evidence F1, and region-type macro F1.

Results

generic: MAE=2.0, item-F1=0.75, evidence-F1=0.5, kappa=0.1356
no_graph: MAE=1.6667, item-F1=0.7273, evidence-F1=0.5, kappa=0.2
scoremap: MAE=1.3333, item-F1=0.7647, evidence-F1=0.5417, kappa=0.3878

Ablations and Failure Cases

Removing type-specific routing hurts rubric-item alignment most strongly on mixed answers containing code and prose. Removing graph constraints mainly harms ordered algorithm-step questions and Gantt-chart traces. Failure cases include unusually terse answers, unsupported diagram families, and semantically correct wording that does not share enough lexical overlap with the rubric.

Limitations and Ethics

The current prototype uses lightweight region inputs and synthetic handwritten-style pages for the included benchmark. It should be treated as an auditable second-opinion grader, not a replacement for human judgment. Real deployment would require stronger transcription, fairness testing across handwriting styles, and privacy-aware data collection.

References

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